Xueshen Liu

(+1) 734-546-5139 | liuxs@umich.edu | xenshinu.github.io | github.com/xenshinu | Google Scholar

Summary

Final-year Ph.D. candidate in CSE at the University of Michigan (advisor: Prof. Z. Morley Mao), graduating by April 2027. I build **elastic** and **heterogeneous** distributed systems for efficient **LLM** training, inference, and RL.

Education

University of Michigan — Ann Arbor, MI
Ph.D. & B.S. in Computer Science and Engineering; GPA 3.9/4.0 — *Aug. 2020 – Apr. 2027 (exp.)*

Shanghai Jiao Tong University — Shanghai, China
B.S. in Electrical and Computer Engineering; GPA 3.7/4.0; Outstanding Graduate — *Aug. 2018 – Aug. 2022*

Experience

Citadel Securities, Systematic Equity — Miami, FL
Quantitative Researcher Intern — *Jun. 2026 – Aug. 2026*
- Built full-stack ML infrastructure: TPU/GPU kernels, training, and inference.
- Drove performance gains through end-to-end workload and hardware research.

Google, System Research @ Google — Seattle, WA
Student Researcher — *May 2025 – Dec. 2025*
- Profiled the LLM RL pipeline; rollout is a dominant yet elastic stage that fits small, dynamic instances.
- Built **RLBoost** on GCP, harvesting fragmented spot resources to cut RL training cost and raise cloud utilization.

General Motors, Connected Autonomous Vehicle Lab — Warren, MI
Research Intern — *May 2024 – Aug. 2024*
- Developed a parallel **deep factor graph** model that handles delayed perception data with real-time responsiveness.

University of Michigan, CSE 589 Advanced Computer Networks — Ann Arbor, MI
Graduate Student Instructor — *Sept. 2024 – Dec. 2024*
- Led discussions and office hours, lectured on distributed SDN, and mentored graduate research projects.

Selected Publications

Foundry: Template-Based CUDA Graph Context Materialization for Fast LLM Serving Cold Start — *In Submission*
- Materializes CUDA graphs with their **execution context** for fast restore; graph sharing enables **auto-scaling** DP/EP/TP.

RLBoost: Harvesting Preemptible Resources for Cost-Efficient RL on LLMs — *NSDI'26*
- Offloads rollout to preemptible instances via token-level tracking and pull-based weight sync, cutting cost by up to **49%**.

Dr. Post-Training: A Data Regularization Perspective on LLM Post-Training — *In Submission*
- Uses general-domain data as **regularization** to reduce overfitting when adapting LLMs with limited target data.

HeterMoE: Efficient Training of Mixture-of-Experts Models on Heterogeneous GPUs — *In Submission*
- Disaggregates MoE models and **assigns experts to older GPUs** (V100, T4) to maximize utilization and reduce cost.

CAKE: Compute Or Load KV Cache? Why Not Both? — *ICML'25*
- Cuts long-context prefill latency with **bidirectional KV cache generation** that overlaps compute and I/O.

Plato: Plan to Efficiently Decode for Large Language Model Inference — *COLM'25*
- Decomposes questions into a **dependency graph** of sub-problems and decodes them in parallel with shared context.

LTE: Learn-To-be-Efficient: Build Structured Sparsity in Large Language Models — *NeurIPS'24 Spotlight*
- Trains LLMs for structured activation sparsity; a Triton **gather-scatter MLP kernel** gives speedup linear in sparsity.

mm2-gb: GPU Accelerated Minimap2 for Long Read DNA Mapping — *ACM BCB'24 Oral*
- Accelerates irregular chaining on AMD GPUs (HIP) with **split kernels** and prioritized **persistent-kernel** scheduling.

Honors, Talks & Service

- **Awards:** UMich Doctoral Intern Fellowship (2026); Roger King Scholarship (2021); RoboMaster Runner-up (2019)
- **Invited Talks:** NVIDIA Dynamo (Oct. 2026, RLBoost & Foundry); Amazon SFAI (2026, RLBoost & Foundry); General Motors (2024); AMD (2024, mm2-gb)
- **Reviewer:** COLM'26, ICLR'26, ICLR'25, COLING'25

Skills

ML Systems: VeRL, PyTorch, JAX/Pallas, CUTLASS, DeepSpeed, NCCL, SGLang, vLLM, FlashAttention
Languages: Python, Rust, Triton, CUDA, HIP, C/C++, Go
Tools: Kubernetes, Nsight Systems/Compute, XProf, Perfetto, Slurm, Docker, Git, Claude Code/Codex